

Reproducing Kernel Hilbert Space

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Kernel methods

- ▶ Kernel smoothing is a technique used for non-parametric curve estimation.
- ▶ The quality of the estimated curve depends on
 - ▶ the bandwidth parameter (smoothness), and on
 - ▶ the kernel function (weight).
- ▶ For the kernel, the key parameter is its order which is related both
 - ▶ to the number of vanishing moments, and
 - ▶ to the number of existing derivatives of the underlying curve.

Vector space

A set $\mathbf{V}$ closed under finite vector addition and scalar multiplication. The scalars are members of a field F . $\mathbf{V}$ is a vector space over F . $\mathbb{R}^n$ is a real vector space, whereas $\mathbb{C}^n$ is a complex vector space. For a set $\mathbf{V}$ to be a vector space, the following conditions must hold $\forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbf{V}$ and $\forall a, b \in F$:

- ▶ Commutativity: $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
- ▶ Associativity: $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
- ▶ Additive identity: $\forall \mathbf{x}, \mathbf{0} + \mathbf{x} = \mathbf{x} + \mathbf{0} = \mathbf{x}$
- ▶ Additive inverse: $\forall \mathbf{x}, \exists -\mathbf{x}$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
- ▶ Associativity: $a(b\mathbf{x}) = (ab)\mathbf{x}$
- ▶ Distributivity of scalar sums: $(a + b)\mathbf{x} = a\mathbf{x} + b\mathbf{x}$
- ▶ Distributivity of vector sums: $a(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + a\mathbf{y}$
- ▶ Scalar multiplication identity: $1\mathbf{x} = \mathbf{x}$

Hilbert space

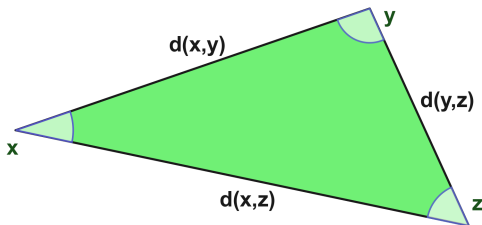
A Hilbert space $\mathcal{H}$ is a real (complex) inner product vector space that is also a complete metric space with respect to the distance function induced by the inner product. This inner product $\langle \cdot, \cdot \rangle$ associates a real (complex) number to each pair of elements $\langle \mathbf{x}, \mathbf{y} \rangle$ of $\mathcal{H}$, and satisfies the following properties:

- ▶ $\langle \mathbf{y}, \mathbf{x} \rangle = \overline{\langle \mathbf{x}, \mathbf{y} \rangle}$
- ▶ $\langle a\mathbf{x}_1 + b\mathbf{x}_2, \mathbf{y} \rangle = \langle a\mathbf{x}_1, \mathbf{y} \rangle + \langle b\mathbf{x}_2, \mathbf{y} \rangle$
- ▶ $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0, \langle \mathbf{x}, \mathbf{x} \rangle = 0 \iff \mathbf{x} = \mathbf{0}$
- ▶ $\langle \mathbf{x}, a\mathbf{y}_1 + b\mathbf{y}_2 \rangle = \bar{a}\langle \mathbf{x}, \mathbf{y}_1 \rangle + \bar{b}\langle \mathbf{x}, \mathbf{y}_2 \rangle$

$\mathcal{H}$ is a real vector space and the inner product yields real (complex) values, and is bilinear, i.e. (id est), linear in each argument.

Distance in Hilbert spaces

The norm is a real valued function defined as $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$, and the distance between $\mathbf{x}, \mathbf{y} \in \mathcal{H}$ is given by $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$. This distance function obeys the triangle inequality, i.e., $d(\mathbf{x}, \mathbf{z}) \leq d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z})$.



This property is a consequence of the Cauchy–Schwarz inequality $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$.

Completeness of Hilbert spaces

$\mathcal{H}$ is said complete if every Cauchy sequence of vectors

$$\{\mathbf{u}_0, \mathbf{u}_1, \dots\}$$

converges with respect to the norm $\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}$ to an element of $\mathcal{H}$. If a series of vectors converges absolutely in the sense that of

$$\sum_{k=0}^{\infty} \|\mathbf{u}_k\| < \infty$$

then the series converges in $\mathcal{H}$, in the sense that the partial sums

$$\mathbf{s}_m = \sum_{k=0}^m \mathbf{u}_k$$

converge to an element of $\mathcal{H}$, i.e. $\mathbf{s}_m \in \mathcal{H}$

RKHSs overview

- ▶ Let $\mathcal{X}$ represent a nonempty set (the input space), and g represent a real-valued function

$$g : \mathcal{X} \rightarrow \mathbb{R}.$$

- ▶ Let $\mathcal{H}$ represent an hypothesis space (a Hilbert space) with well defined pointwise functions $\forall x \in \mathcal{X}$.
- ▶ The pointwise evaluators $g \rightarrow g(x)$ are considered linear and bounded over $\mathcal{H}$, i.e. $\forall x \in \mathcal{X}$ there exists $C_x < \infty$ such that

$$|g(x)| \leq C_x \|g\|_{\mathcal{H}}, \forall g \in \mathcal{H}$$

The goal is to estimate maps for make predictions over the whole $\mathcal{X}$.

Definitions

- ▶ A reproducing kernel Hilbert space (RKHS) $\mathcal{H}$ over $\mathcal{X}$ is a space containing functions $g : \mathcal{X} \rightarrow \mathbb{R}$ where

$$|g(x)| \leq C_x \|g\|_{\mathcal{H}}, \forall g \in \mathcal{H}$$

holds.

- ▶ A symmetric function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is called positive definite kernel if $\forall p \in \mathbb{Z}$

$$\sum_{i=1}^p \sum_{j=1}^p c_i c_j K(x_i, x_j) \geq 0, \forall (x_k, c_k) \in (\mathcal{X}, \mathbb{R})$$

The kernel section $\mathcal{K}_x$ centered at x is the function from $\mathcal{X}$ to $\mathbb{R}$ defined by $\mathcal{K}_x(a) = K(a, x), \forall a \in \mathcal{X}$.

Theorems

- ▶ For every RKHS $\mathcal{H}$ of real-valued functions defined over $\mathcal{X}$ there exists a unique positive definite kernel $\mathcal{K}$ such that

$$\langle \mathcal{K}_x, g \rangle = g(x), \forall (x, g) \in (\mathcal{X}, \mathcal{H})$$

- ▶ For every positive definite kernel $\mathcal{K}$, there exists a unique RKHS $\mathcal{H}$ of real-valued functions defined over $\mathcal{X}$ such that

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Uniqueness

- ▶ A RKHS $\mathcal{H}$ of real-valued functions defined over $\mathcal{X}$ is completely defined by a kernel $\mathcal{K}$.
- ▶ A RKHS $\mathcal{H}$ is generated by the kernel sections $\mathcal{K}_x$.
- ▶ Let $\mathbf{S}$ denote the subspace spanned by $\{\mathcal{K}_x\}_{x \in \mathcal{X}}$ and for any $g \in \mathbf{S}$, say $g = \sum_{i=1}^p c_i \mathcal{K}_{x_i}$ define the norm

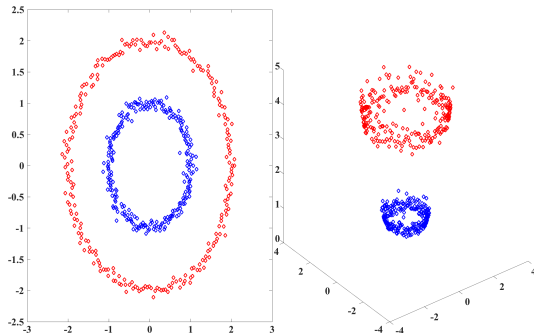
$$\|g\|_{\mathcal{H}}^2 = \sum_{i=1}^p \sum_{j=1}^p c_i c_j \mathcal{K}(x_i, x_j).$$

$\mathcal{H}$ is the union of $\mathbf{S}$ and all the limits with respect to $\|\cdot\|_{\mathcal{H}}^2$ of the Cauchy sequences contained in $\mathbf{S}$.

- ▶ A consequence of this construction is that any $g \in \mathcal{H}$ inherits the kernel properties.

The kernel trick

The kernel $\mathcal{K}(x_i, x_j)$ transforms datasets to a higher-dimension space $\mathbb{R}^M$ (possibly infinite). By using a kernel $\mathcal{K}(x_i, x_j)$ one may determine nonlinear decision boundaries.



$$[x_1, x_2] \in \mathbb{R}^2 \rightarrow [x_1, x_2, x_3], \in \mathbb{R}^3, x_3 = x_1^2 + x_2^2$$

Least squares regularization

Given (x_i, y_i) , $i = 1, \dots, N$, where x_i is the input and y_i is the corresponding output measurement. The goal is to find a function with good prediction capability for future data. For a new pair (x, y) , the prediction $y \approx f(x)$.

$$V_N(f) = \frac{1}{N} \sum_{i=1}^N [y_i - f(x_i)]^2 + \gamma \|f\|_{\mathcal{H}}^2, \hat{g} = \underset{f \in \mathcal{H}}{\operatorname{argmin}} V_N(f),$$

where $\mathcal{H}$ denotes a RKHS with norm $\|\cdot\|_{\mathcal{H}}^2$. $\hat{g}$ minimizes an objective function consisting of two terms.

- ▶ the quadratic loss term - adherence to experimental data
- ▶ the regularizer term - well-posedness - ill-conditioning

The representer theorem

- ▶ The solution of $\hat{g} = \operatorname{argmin}_{f \in \mathcal{H}} V_N(f)$ is unique and given by

$$\hat{g} = \sum_{i=1}^N \hat{c}_i \mathcal{K}_{x_i}$$

where the scalars $\hat{c}_i$ are the components of the vector

$$\hat{c} = (\mathbf{K} + \gamma \mathbf{I}_N)^{-1} \mathbf{Y}$$

$\mathbf{Y}$ is the column vector with i -th element y_i , $\mathbf{I}_N$ is the $N \times N$ identity matrix and the (i, j) entry of $\mathbf{K}$ is $\mathcal{K}(x_i, x_j)$.

Regularized least squares

$$V(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2$$

$$\hat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^d} V(\mathbf{x})$$

The solutions is unique if

$$\forall \lambda > 0, \lambda \in \mathbb{R},$$

$$\frac{d}{dt} \{V(\hat{\mathbf{x}} + t\mathbf{w})\} \big|_{t=0} = 0,$$

$$t \in \mathbb{R}, \forall \mathbf{w} \in \mathbb{R}^d.$$

Expanding $V(\hat{\mathbf{x}} + t\mathbf{w})$

$$\frac{d}{dt} \left\{ \|\mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b}\|^2 + \lambda \|(\hat{\mathbf{x}} + t\mathbf{w})\|^2 \right\} |_{t=0} = 0$$

$$\|\mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b}\|^2 = \langle \mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b}, \mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b} \rangle$$

$$\lambda \|\hat{\mathbf{x}} + t\mathbf{w}\|^2 = \lambda \langle \hat{\mathbf{x}} + t\mathbf{w}, \hat{\mathbf{x}} + t\mathbf{w} \rangle$$

$$\langle \mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b}, \mathbf{A}(\hat{\mathbf{x}} + t\mathbf{w}) - \mathbf{b} \rangle = \langle \mathbf{A}\hat{\mathbf{x}} - \mathbf{b} + t\mathbf{A}\mathbf{w}, \mathbf{A}\hat{\mathbf{x}} - \mathbf{b} + t\mathbf{A}\mathbf{w} \rangle$$

$$\langle \cdot, \cdot \rangle = \|\mathbf{A}\hat{\mathbf{x}}\|^2 + 2t \langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\mathbf{w} \rangle + t^2 \|\mathbf{A}\mathbf{w}\|^2 - 2t \langle \mathbf{b}, \mathbf{A}\mathbf{w} \rangle - 2 \langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{b} \rangle + \|\mathbf{b}\|^2$$

$$\langle \hat{\mathbf{x}} + t\mathbf{w}, \hat{\mathbf{x}} + t\mathbf{w} \rangle = \|\hat{\mathbf{x}}\|^2 + 2t \langle \hat{\mathbf{x}}, \mathbf{w} \rangle + t^2 \|\mathbf{w}\|^2$$

$$\frac{d}{dt} \{ V(\hat{\mathbf{x}} + t\mathbf{w}) \} |_{t=0} = 2 \langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\mathbf{w} \rangle - 2 \langle \mathbf{b}, \mathbf{A}\mathbf{w} \rangle + \lambda 2 \langle \hat{\mathbf{x}}, \mathbf{w} \rangle$$

Taking the transposes ...

$$\langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\mathbf{w} \rangle - \langle \mathbf{b}, \mathbf{A}\mathbf{w} \rangle + \lambda \langle \hat{\mathbf{x}}, \mathbf{w} \rangle = 0$$

$$\langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\mathbf{w} \rangle = (\mathbf{A}\hat{\mathbf{x}})^\top \mathbf{A}\mathbf{w} = \left(\mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} \right)^\top \mathbf{w} = \left\langle \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}}, \mathbf{w} \right\rangle$$

$$\langle \mathbf{b}, \mathbf{A}\mathbf{w} \rangle = \langle \mathbf{A}\mathbf{w}, \mathbf{b} \rangle = (\mathbf{A}\mathbf{w})^\top \mathbf{b} = \mathbf{w}^\top \mathbf{A}^\top \mathbf{b} = \left\langle \mathbf{w}, \mathbf{A}^\top \mathbf{b} \right\rangle = \left\langle \mathbf{A}^\top \mathbf{b}, \mathbf{w} \right\rangle$$

$$\langle \mathbf{A}\hat{\mathbf{x}}, \mathbf{A}\mathbf{w} \rangle - \langle \mathbf{b}, \mathbf{A}\mathbf{w} \rangle = \left\langle \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}}, \mathbf{w} \right\rangle - \left\langle \mathbf{A}^\top \mathbf{b}, \mathbf{w} \right\rangle = \left\langle \mathbf{A}^\top \mathbf{A}\hat{\mathbf{x}} - \mathbf{A}^\top \mathbf{b}, \mathbf{w} \right\rangle$$

$$\left\langle \left(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I} \right) \hat{\mathbf{x}} - \mathbf{A}^\top \mathbf{b}, \mathbf{w} \right\rangle = 0, \forall \mathbf{w} \in \mathbb{R}^d$$

$$\left(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I} \right) \hat{\mathbf{x}} = \mathbf{A}^\top \mathbf{b}$$

$$\hat{\mathbf{x}} = \left(\mathbf{A}^\top \mathbf{A} + \lambda \mathbf{I} \right)^{-1} \mathbf{A}^\top \mathbf{b}$$

Symmetric positive definite kernels

- ▶ Linear

$$\mathcal{K}(x, y) = \langle x, x \rangle$$

- ▶ Gaussian

$$\mathcal{K}(x, y) = e^{-\frac{|x-y|^2}{\sigma^2}}, \sigma > 0$$

- ▶ Laplacian

$$\mathcal{K}(x, y) = e^{-\frac{|x-y|}{\eta}}, \eta > 0$$

- ▶ Polynomial

$$\mathcal{K}(x, y) = (\langle x, y \rangle + 1)^d, d \in \mathbb{Z}$$

RKHSs of linear dynamic systems

- ▶ Discrete-time: the system input is $u : \mathcal{Z} \rightarrow \mathbb{R}$, and any input vector $x_t \in \mathcal{X}$ is indexed by $t \in \mathcal{Z}$, i.e.,

$$x_t = [u_t \quad u_{t-1} \quad \cdots \quad u_{t-(m-1)}]^T$$

where m is the system memory, and $\mathcal{X} \subset \mathbb{R}^m$.

- ▶ Continuous-time: the system input is $u : \mathbb{R} \rightarrow \mathbb{R}$, and any input vector $x_t : \mathbb{R}_+ \rightarrow \mathbb{R}$ is defined by

$$x_t(\tau) = u(t - \tau), \tau \geq 0,$$

i.e., x_t contains all past inputs up to the instant t .

Impulse response

For linear time invariant system, the relationship between input and output is given by

$$y(t_i) = \int_0^{\infty} u(t_i - s) h(s) ds + e(t_i), i = 1, \dots, N$$

where $e(t_i)$ is white noise and the unknown function h is now the system impulse response.

- ▶ The domain of the function h is one-dimensional and given by the positive real axis, i.e. $\mathcal{X} = \mathbb{R}^+$ and $h : \mathbb{R}^+ \rightarrow \mathbb{R}$.
- ▶ The measurement model involves the functional $L_i[\cdot]$ given by

$$L_i[h] = \int_0^{\infty} u(t_i - s) h(s) ds$$

Nonparametric identification

To determine h based on input u and $\mathbf{Y}$, where $y_i = y(t_i)$ and $\mathbf{Y} = [y_1 \ y_2 \ \cdots \ y_N]^T$. In this case the objective function is

$$V_N(h) = \frac{1}{N} \sum_{i=1}^N (y_i - L_i[h])^2 + \gamma \|h\|_{\mathcal{P}}^2, \hat{g} = \underset{h \in \mathcal{P}}{\operatorname{argmin}} V_N(h)$$

where $\mathcal{P}$ is the causal RKHSs of impulse responses. To determine $\hat{g}$ one needs to define the output kernel matrix $\mathbf{O} \in \mathbb{R}^{N \times N}$, i.e.

$$O(t, \tau) = \int_0^\infty u(t-x) \left(\int_0^\infty u(\tau-a) K(x, a) da \right) dx$$

The (i, j) entry of $\mathbf{O}$ is given by $\mathbf{O}_{ij} = L_i[L_j[K]] = O(t_i, t_j)$

Impulse response kernels

For obtaining a finite-dimensional representation $L_i[h]$ must be continuous on $\mathcal{H}$, i.e.,

$$\forall i \quad \exists C_i < \infty : |L_i[h]| \leq C_i \|h\|_{\mathcal{H}}, \forall h \in \mathcal{H}$$

The linear functional $L_i[\cdot]$ is continuous if and only if $L_i[\mathcal{K}_x]$ is a function in $\mathcal{H}$.

- ▶ If $\mathcal{H}$ is a RKHS induced by $\mathcal{K}$ and each $L_i : \mathcal{H} \rightarrow \mathbb{R}$ is continuous, then $\hat{g} = \operatorname{argmin}_{h \in \mathcal{P}} V_N(h)$ is given by

$$\hat{g}(x) = \sum_{i=1}^N \hat{c}_i L_i[\mathcal{K}_x]$$

where the scalars $\hat{c}_i$ are the components of the vector

$$\hat{\mathbf{c}} = (\mathbf{O} + \gamma \mathbf{I}_N)^{-1} \mathbf{Y}.$$